

4.1 Entanglement temperatures

Let a general multi-interval region in $d = 2$ (at the line $t = 0$) be $A = \cup_{i=1}^N (l_i, r_i)$. In a CFT the modular Hamiltonian K of the vacuum is non local except for the case of a single interval. However, it was conjectured in [9, 10] that for multi-intervals there is a local term in K having the same universal structure for any CFT. This writes

$$K_{\text{loc}} = \int dx \beta(x) T_{00}(x), \quad (4.1)$$

where the “local inverse temperatures” are

$$\beta(x) = \frac{2\pi}{\sum_{i=1}^n \left(\frac{1}{x-l_i} - \frac{1}{x-r_i} \right)}. \quad (4.2)$$

The modular Hamiltonian for multi-interval regions in $d = 2$ is known explicitly for free massless scalar and fermions [24, 25], and the local term agrees with (4.1).

A clear operational definition of these entanglement temperatures is that they measure the ratio between the expectation value of the modular Hamiltonian and the energy for very localized high energy excitations:

$$\beta(x) = \lim_{E \rightarrow \infty, \Delta x \rightarrow 0} \frac{\langle \psi | K | \psi \rangle}{\langle \psi | H | \psi \rangle}. \quad (4.3)$$

Here $|\psi\rangle$ is an excitation above the vacuum localized in Δx around the point x and has energy E . This limit can be argued to exist based on monotonicity of the relative entropy [9].

We will now show the universality of these entanglement temperatures in a more formal way. To test the entanglement temperatures we will be using a chiral field operator acting on the vacuum as the excited state probe. We would like to compute (for a real chiral field ϕ)

$$\beta(x) = \lim_{y \rightarrow x} \frac{\langle 0 | \phi(x) K \phi(y) | 0 \rangle}{\langle 0 | \phi(x) H \phi(y) | 0 \rangle}. \quad (4.4)$$

This measures the ratio between the modular energy K and the ordinary energy H . As a regularization we set $x \neq y$ and take the limit $x \rightarrow y$.⁷ The modular operator K can be obtained from the modular flow $\Delta^{i\tau} = e^{-iK\tau}$ with $K = -\log \Delta$. We get the entanglement temperature as a limit of a modular evolved correlator

$$\beta(x) = \lim_{y \rightarrow x} \lim_{\tau \rightarrow 0} - \frac{\partial_\tau \langle 0 | \phi(x) \Delta^{i\tau} \phi(y) | 0 \rangle}{\partial_{y^0} \langle 0 | \phi(x) \phi(y) | 0 \rangle}. \quad (4.5)$$

For a general chiral field ϕ in a CFT, S. Hollands obtained a general structure of the modular evolved correlator [26]. From a clever use of the KMS condition for the modular flow and analyticity in the Euclidean plane Hollands maps the problem into a Riemann-Hilbert problem in the plane with a cut at A . His expression will be enough to get the entanglement temperatures by the limit (4.5).

Take a chiral field ϕ of dimension h . By locality h is a half integer number. The field is normalized such that

$$\langle 0 | \phi(x) \phi(y) | 0 \rangle = \frac{e^{-i\pi h}}{(x - y - i0^+)^{-2h}}. \quad (4.6)$$

Defining

$$\Pi_l(x) = \prod_{j=1}^N (x - l_j), \quad \Pi_r(x) \equiv \prod_{i=1}^N (x - r_i), \quad Z(x) = \frac{1}{2\pi} \ln(-\Pi_l(x)/\Pi_r(x)), \quad (4.7)$$

⁷The structure of the singularity turns out to be the same in numerator and denominator, so we can equivalently take fields smeared in a short interval around a point.